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1. A retail company's nightly financial reconciliation fails due to data inconsistency from multiple sources. As the lead data engineer, which of the following is the most robust and scalable long-term solution? 1/1 point
- ☐ Create a real-time streaming pipeline to process and reconcile each transaction the moment it occurs.
 - ☐ Load all raw data directly into the data warehouse and perform all cleaning using the warehouse's native SQL capabilities.
 - ☐ Develop a series of ad-hoc scripts to pull and clean data, which can be manually triggered each night.
 - ☐ Design an automated, end-to-end batch data pipeline that orchestrates the collection, cleansing, and validation of data on a nightly schedule.
 - ☒ **Correct**
This provides a scalable, reliable, and automated solution that is purpose-built for handling large, periodic data integration tasks and addresses the root cause of the failures.
2. An analytics team needs to prepare a massive dataset containing five years of historical sales data to train a customer trend prediction model. Why is batch processing the most suitable approach for this task? 1/1 point
- ☐ Because batch processing is the only method that can load data into a data warehouse.
 - ☐ Because batch processing requires less data cleansing and preparation than other methods.
 - ☐ Because batch processing is designed to efficiently handle very large, bounded datasets, making it ideal for this type of historical data preparation.
 - ☐ Because batch processing provides real-time, up-to-the-minute data for model training.
 - ☒ **Correct**
Correct! ML model training for trend analysis requires large volumes of historical data. Batch pipelines are designed to efficiently process these massive, bounded datasets.
3. Landing raw data in a persistent storage layer before processing is a critical architectural best practice. What is the primary reason this approach builds a more resilient data pipeline? 1/1 point
- ☐ Because cloud storage automatically cleans and validates the data upon upload.
 - ☐ Because it decouples ingestion from processing, allowing the transformation job to be re-run from the raw source data if it fails.
 - ☐ Because it is the only way to secure the data using modern encryption features.
 - ☐ Because uploading to cloud storage is faster than processing the data directly from the source.
 - ☒ **Correct**
Correct! Storing the raw data in a durable location creates a recovery point. If a downstream job fails or has a bug, the process can be safely re-run using the pristine source data.
4. A data engineer is building a new batch pipeline but anticipates a strong possibility of needing to process data in real-time in the future. Which design choice would best "future-proof" the pipeline? 1/1 point
- ☐ Build two separate pipelines now—one for batch and one for streaming—to be prepared for the future.
 - ☐ Choose the most performant batch processing engine, as that will make it easier to adapt to streaming later.
 - ☐ Focus only on the current batch requirements and address the streaming needs in a separate project later.
 - ☐ Select a programming model that is unified for both batch and streaming, allowing the core business logic to be reused.
 - ☒ **Correct**
Correct! A unified model allows logic to be written once and executed on either a batch or streaming runner, making the transition from batch to real-time seamless and efficient.
5. A data engineering team with years of experience developing jobs in Apache Spark wants to migrate to a serverless cloud environment to eliminate infrastructure management. What is the most logical and efficient migration strategy? 1/1 point
- ☐ Adopt a managed or serverless service that can run the existing Spark code with minimal modifications.
 - ☐ Continue to manage the clusters manually on virtual machines in the cloud.
 - ☐ Rewrite the jobs using a new, cloud-native programming model to maximize platform integration.
 - ☐ Re-implement the data processing logic using the cloud data warehouse's native SQL engine.
 - ☒ **Correct**
Correct! This strategy allows the team to migrate quickly by leveraging their existing skills and ecosystem, providing the fastest path to value.
6. A company's daily data volume triples during peak sales events, causing the existing pipeline with fixed resource allocations to fail. What is the primary batch data pipeline challenge that this company is facing? 1/1 point
- ☐ Reliability and Observability
 - ☐ Data Quality and Consistency
 - ☐ Complexity and Maintainability
 - ☐ Data Volume and Scalability
 - ☒ **Correct**
Correct! The pipeline's inability to handle a rapid increase in data volume points directly to a challenge of scalability.
7. A batch job failed overnight, but the operations team didn't find out for several hours, delaying critical reports. Engineers had trouble diagnosing the issue because error messages were difficult to find. Which challenge does this represent, and what category of tools is essential for solving it? 1/1 point
- ☐ Data Volume and Scalability, solved by data storage and warehousing services.
 - ☐ Data Quality and Consistency, solved by data transformation frameworks.
 - ☐ Reliability and Observability, solved by centralized logging and metrics-based monitoring services.
 - ☐ Complexity and Maintainability, solved by using a single, monolithic script.
 - ☒ **Correct**
Correct! This is a classic example of poor observability. Centralized logging helps diagnose issues, and monitoring provides alerts to notify teams of failures immediately.
8. A company's CTO wants to reduce operational costs. The current on-premise system runs 24/7, but the daily batch job only takes four hours to complete. How does migrating to a serverless cloud-based batch processing service directly address the CTO's goal? 1/1 point
- ☐ By being latency tolerant, which allows for flexible job scheduling.
 - ☐ By being resource efficient, only charging for compute resources while the job is actively running.
 - ☐ By automatically ensuring all data is complete and accurate.
 - ☐ By processing the data much faster than the on-premise system.
 - ☒ **Correct**
Correct! A key feature of serverless batch processing is resource efficiency. The company would only pay for the computing resources used, eliminating the cost of idle infrastructure.
9. To produce auditable financial statements, a company must perform complex validations across the entire day's sales data at once. What fundamental aspect of batch processing enables this type of comprehensive validation? 1/1 point
- ☐ Its tolerance for latency.
 - ☐ Its high throughput.
 - ☐ Its resource efficiency.
 - ☐ Its ability to operate on a complete, bounded dataset.
 - ☒ **Correct**
Correct! Batch processes are designed to work on complete datasets, which allows for the thorough, cross-record validations that ensure high-data quality for financial audits.
10. A project manager is trying to understand the business impact of using "fully serverless" cloud services. What is the primary benefit that a data engineer would emphasize to this manager? 1/1 point
- ☐ They guarantee that jobs will run faster and more cheaply than on dedicated clusters.
 - ☐ They reduce the total cost of ownership by shifting operational overhead like patching, scaling, and infrastructure management to the cloud provider.
 - ☐ They allow for more granular control over the specific hardware and software versions used.
 - ☐ They provide unlimited and instantaneous scaling with no "cold start" delays.
 - ☒ **Correct**
Correct! The key business value of serverless is reducing operational burden, allowing the engineering team to focus on business logic instead of managing infrastructure.